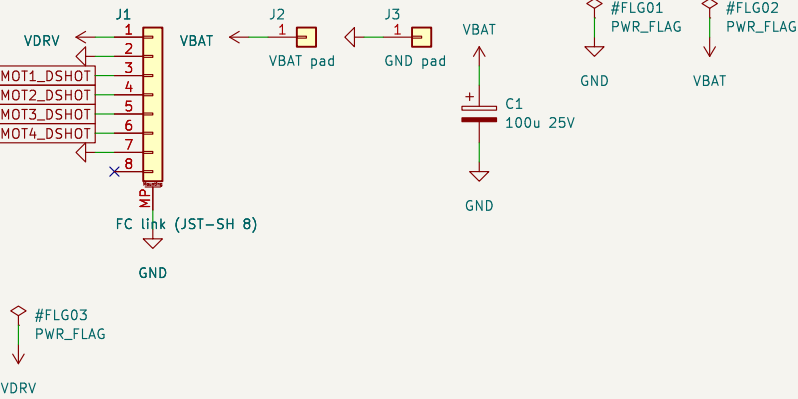


	1	2	3	4	5	6
A	STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2 MicroScout 4-in-1 ESC – root sheet. Sub-sheets: Power and link, Motor 1, Motor 2, Motor 3, Motor 4. Generated by tools/sch from hardware/*/design/*.py – edit the Python, not this file, until G3 hand-off. Connectivity: net labels and power symbols; check report in review/G2/.					A
B	Power and link File: power.kicad_sch	Motor 1 File: motor1.kicad_sch	Motor 2 File: motor2.kicad_sch	Motor 3 File: motor3.kicad_sch		B
C	Motor 4 File: motor4.kicad_sch					C
D					Generated schematic – not reviewed. Values are first-pass G2 choices. STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2 MicroScout (open hardware, CERN-OHL-S-2.0) Sheet: / File: microscout-esc.kicad_sch Title: MicroScout 4-in-1 ESC – root Size: A4 Date: 2026-10-08 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Rev: G2-draft-1 Id: 1/6	D
	1	2	3	4	5	6

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

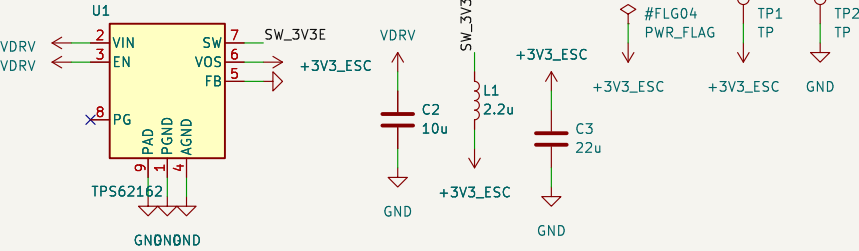
MicroScout 4-in-1 ESC – Power and link

Link to the flight controller (JST-SH 8, mirrors FC J7) and battery wire pads
VDRV (9.6 V, D-054) is switched by the FC's soft switch (D-036); when the drone is off, the drivers and MCUs here are unpowered and the 47 k gate-source resistors hold the FETs off.
Battery current arrives on J2/J3 wire pads from FC J8/J9 (after the FC's current shunt).



ESC logic 3.3 V – TPS62162 from VDRV (D-048; same circuit as the FC 3.3 V rail)

Load: 4x AT32F421 (~25 mA each at 120 MHz, ESTIMATE). VIN 9.6 V (rated 3–17 V). EN tied to VIN; on whenever VDRV is on.



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN-OHL-S-2.0)

Sheet: /Power and link/
File: power.kicad_sch

Title: MicroScout 4-in-1 ESC – Power and link

Size: A3 Date: 2026-10-08

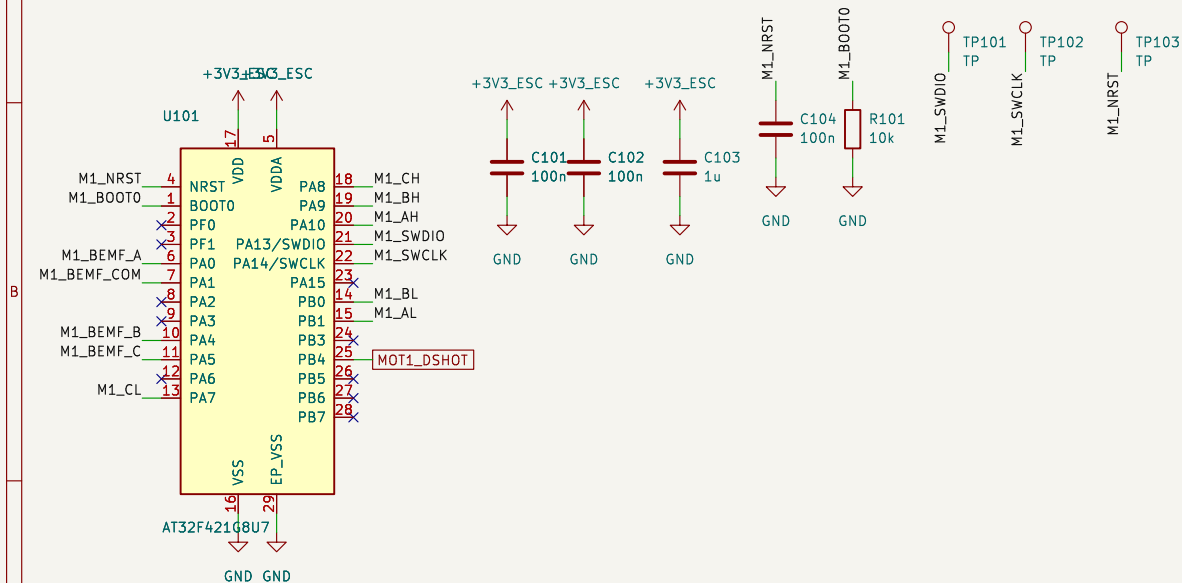
Rev: G2-draft-1

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

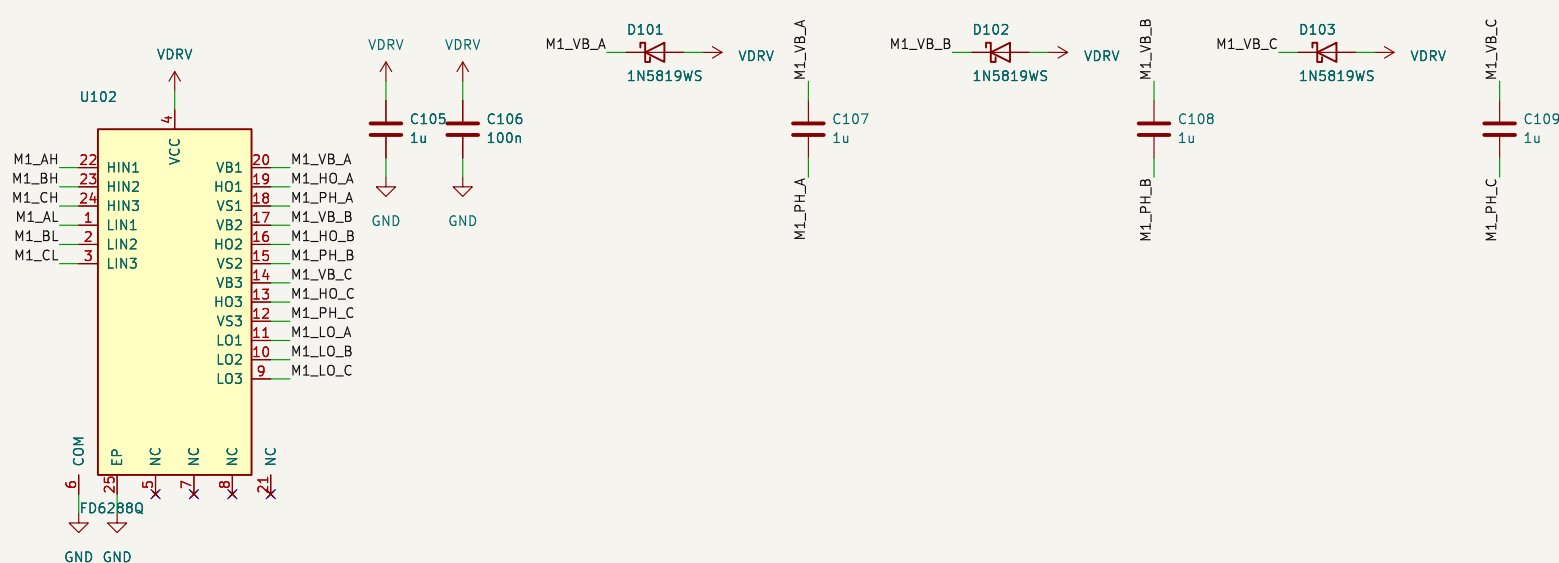
Id: 2/6

MicroScout 4-in-1 ESC – Motor 1

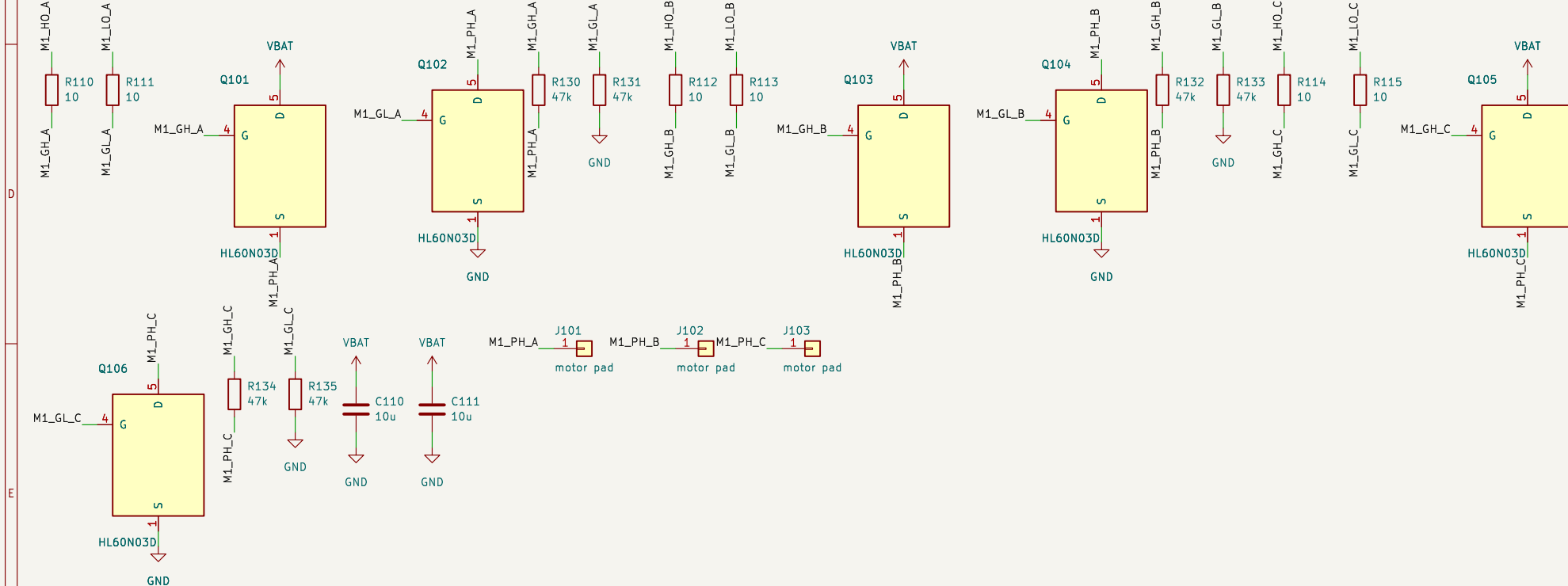
pin assignment of the values UNCONFIRMED). BOOT0 10 k to GND; NRST 100 nF (value UNCONFIRMED).



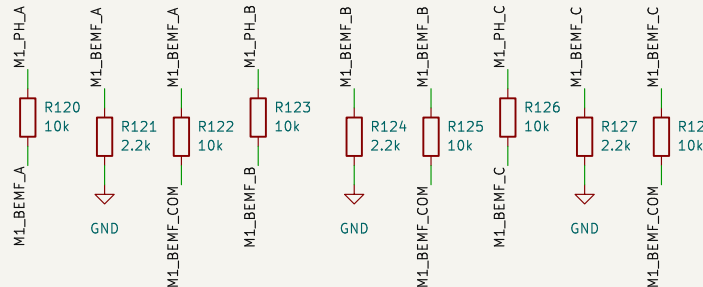
JSMSEMI second source lists 8–20 V VCC (below a 2S pack) – not fitted until its datasheet is read (0Q–13).



the 100 uF polymer bulk capacitor is on the power sheet.



Each sense node: 10 k to the common (virtual neutral) node on PA1.



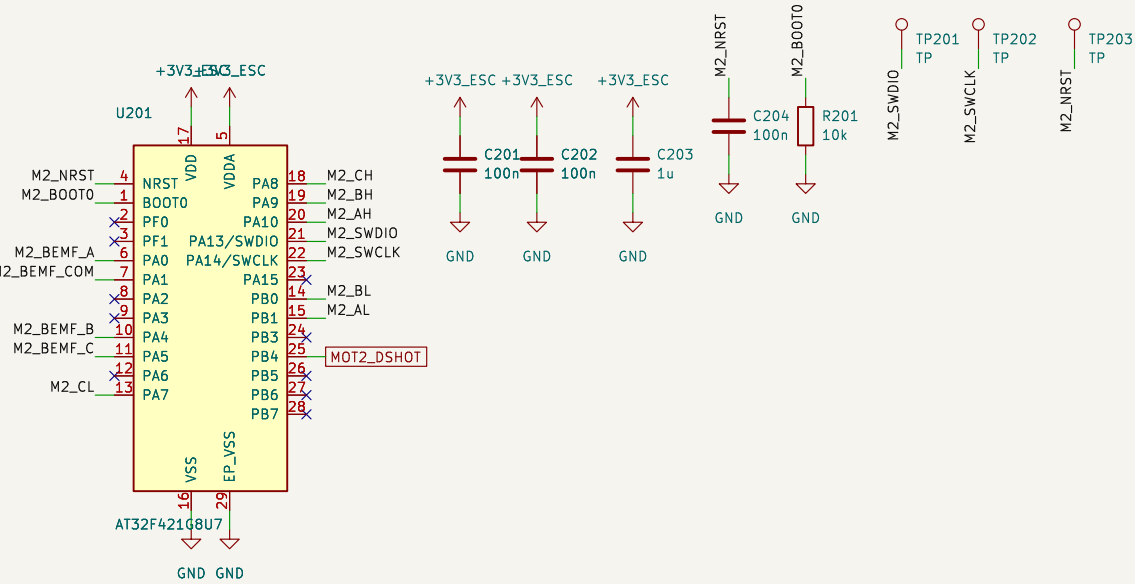
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4
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STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout 4-in-1 ESC – Motor 2

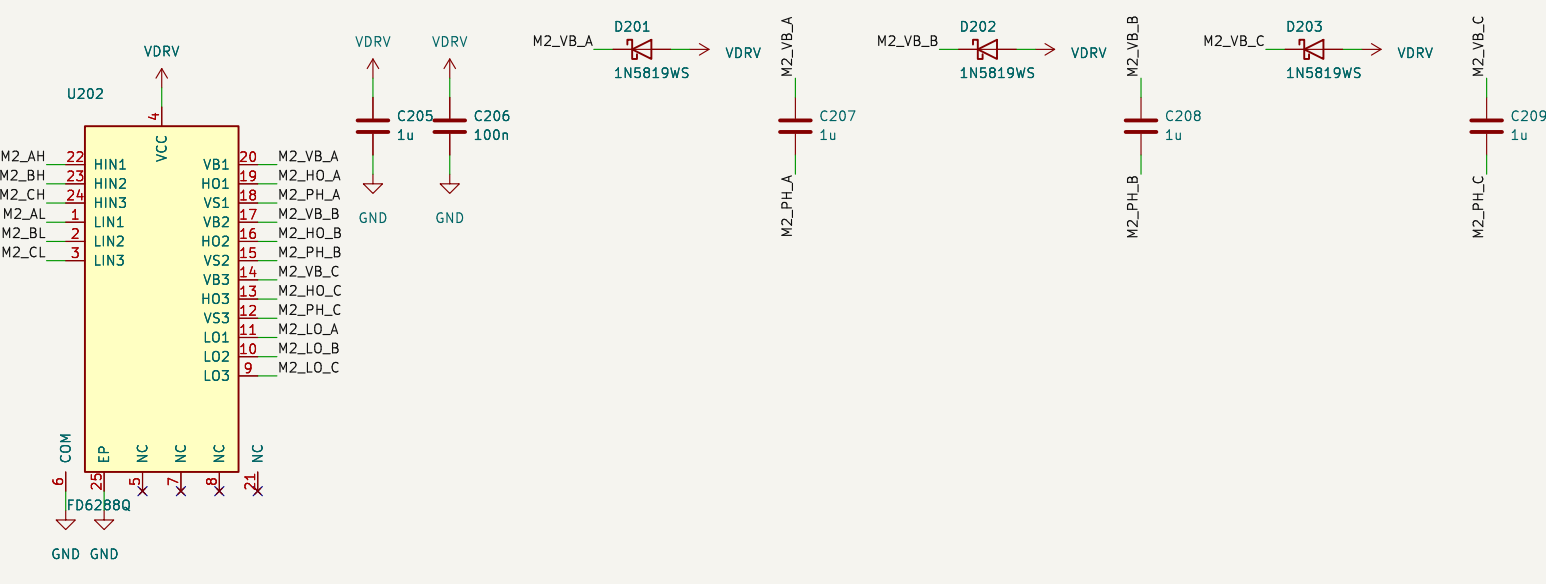
Motor 2 MCU – AT32F421G8U7 with AM32 (pin map: AM32 target OPENESC_20_F421)

PWM: TMR1 CH1–3 + complementary outputs, active high (FD6288 outputs follow inputs; no USE_INVERTED flags).
BEMF comparator inputs PA0/PA4/PA5, neutral PA1. DShot in on PB4 (TMR3 CH1). SWD pads for the first
bootloader flash; later updates over the signal wire. VDD/VDDA decoupling 100 nF each + 1 uF (datasheet Fig. 8;
pin assignment of the values UNCONFIRMED). BOOT0 10 k to GND; NRST 100 nF (value UNCONFIRMED).



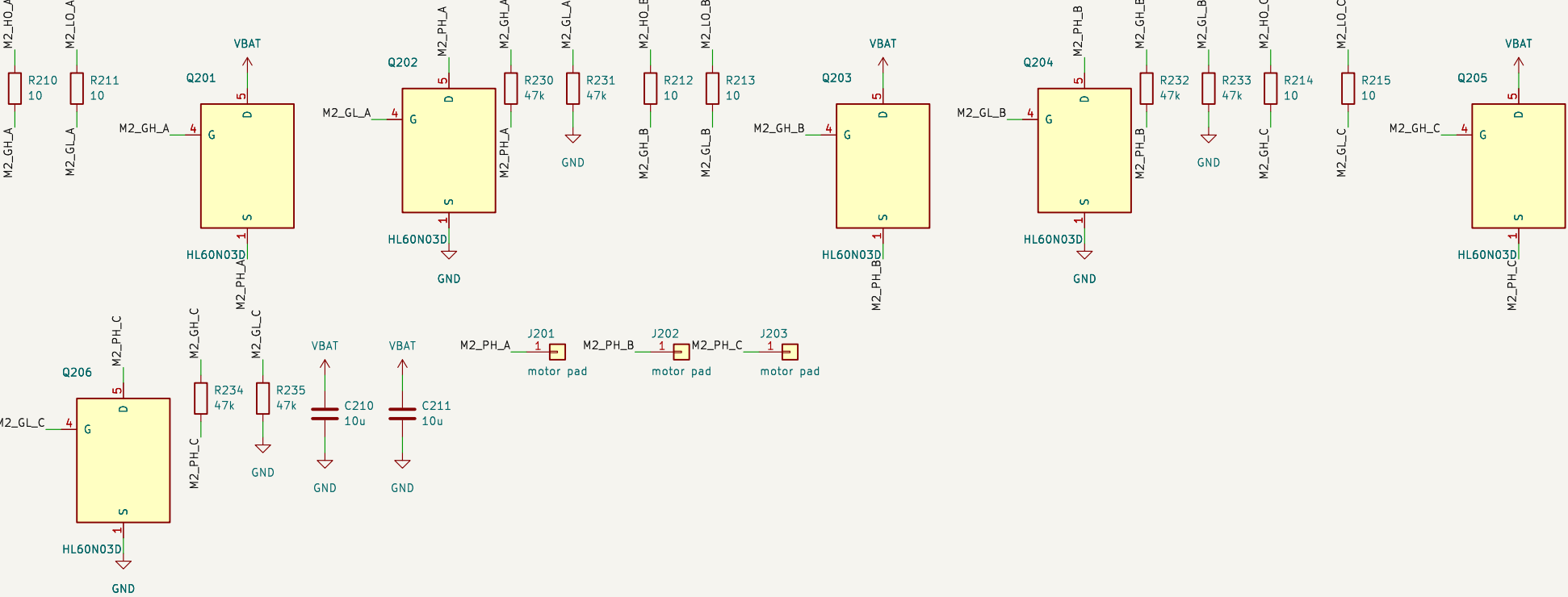
Motor 2 gate driver – FD6288Q (Fortior datasheet V1.6 section 1.5)

VCC = VDRV, 9.6 V from the FC boost (D–054); FD6288 VCC 5–20 V, UVLO 4.6 V typ / 5.0 V max rising.
Bootstrap: external Schottky (FD6288 has none) + 1 uF per phase. Cboot >= Qg / dV = 34 nC / 0.1 V = 0.34 uF;
1 uF gives ~34 mV droop per cycle. 10 Ohm gate resistors: agent choice for edge control, tune at G6.
JSMSEMI second source lists 8–20 V VCC (below a 2S pack) – not fitted until its datasheet is read (OQ–13).



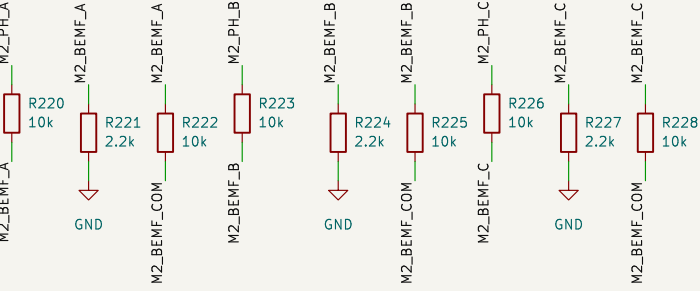
Motor 2 power stage – 6x HL 60N03D (30 V; 4.7 mOhm typ / 6 mOhm max at 10 V gate drive; Qg 34 nC)

Conduction loss at 9.2 A (vendor max current): two FETs conduct, $P = 9.2^2 \times 2 \times 4.7 \text{ mOhm} = 0.80 \text{ W}$ typ (budgets 4c).
1.0 W at 6 mOhm max. 47 k gate–source resistors hold every FET off while the drivers are unpowered (FD6288
datasheet application circuit R2; D–036 keeps VBAT on the bridge when off). 2x 10 uF 25 V across the bridge;
the 100 uF polymer bulk capacitor is on the power sheet.



Motor 2 back–EMF sensing (topology traced from OpenESC–20x20, CERN–OHL–S–2.0;
values scaled for 2S)

Each phase: 10 k to the sense node, 2.2 k to GND → $8.8 \text{ V} \times 2.2 / 12.2 = 1.59 \text{ V}$ max (OpenESC uses 1 k for 6S).
Each sense node: 10 k to the common (virtual neutral) node on PA1.



Generated schematic – not reviewed. Values are first–pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN–OHL–S–2.0)

Sheet: /Motor 2/
File: motor2.kicad_sch

Title: MicroScout 4-in-1 ESC – Motor 2

Size: A3 Date: 2026–10–08
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Rev: G2–draft–1
Id: 4/6

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout 4-in-1 ESC – Motor 3

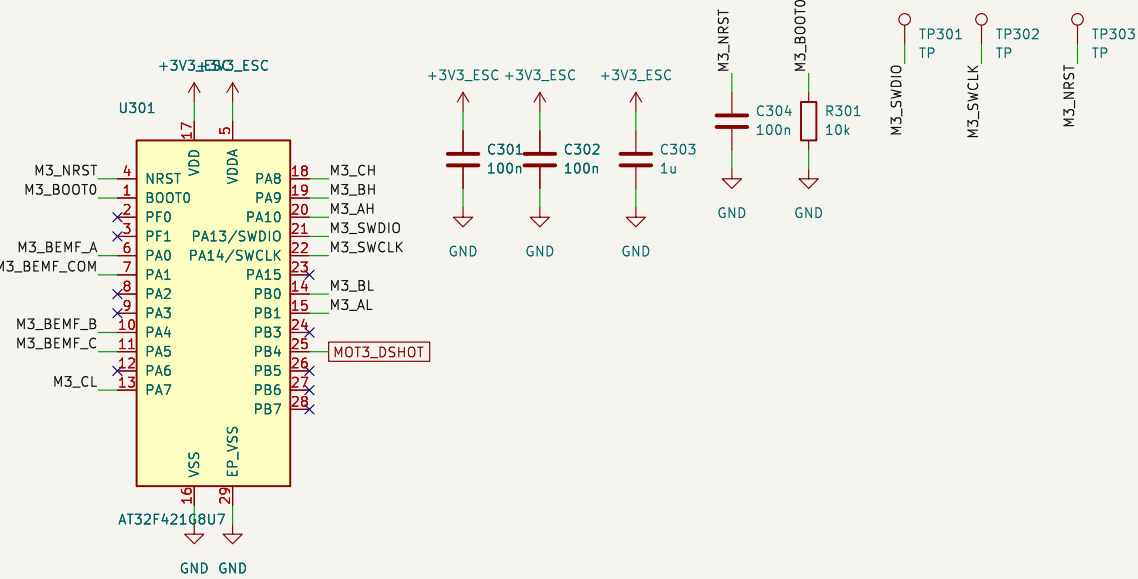
Motor 3 MCU – AT32F421G8U7 with AM32 (pin map: AM32 target OPENESC_20_F421)

PWM: TMR1 CH1–3 + complementary outputs, active high (FD6288 outputs follow inputs; no USE_INVERTED flags).

BEMF comparator inputs PA0/PA4/PA5, neutral PA1. DShot in on PB4 (TMR3 CH1). SWD pads for the first

bootloader flash; later updates over the signal wire. VDD/VDDA decoupling 100 nF each + 1 uF (datasheet Fig. 8;

pin assignment of the values UNCONFIRMED). BOOT0 10 k to GND; NRST 100 nF (value UNCONFIRMED).



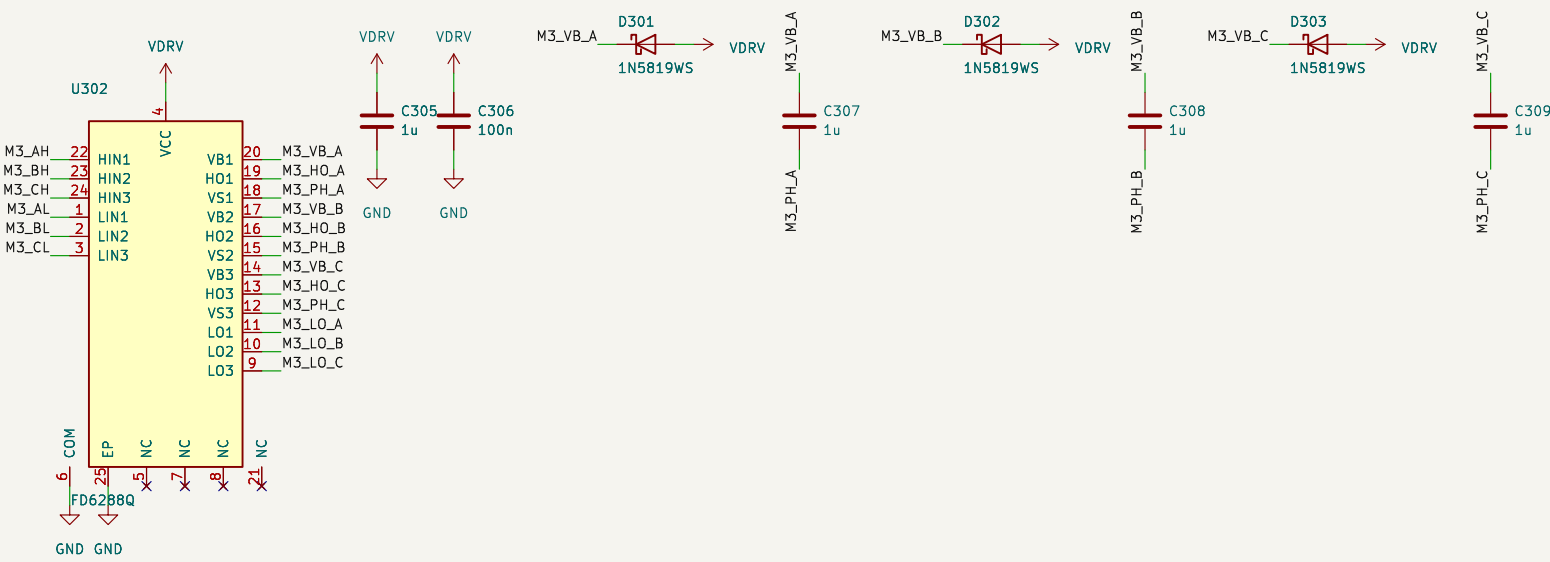
Motor 3 gate driver – FD6288Q (Fortior datasheet V1.6 section 1.5)

VCC = VDRV, 9.6 V from the FC boost (D–054); FD6288 VCC 5–20 V, UVLO 4.6 V typ / 5.0 V max rising.

Bootstrap: external Schottky (FD6288 has none) + 1 uF per phase. Cboot >= Qg / dV = 34 nC / 0.1 V = 0.34 uF;

1 uF gives ~34 mV droop per cycle. 10 Ohm gate resistors: agent choice for edge control, tune at G6.

JSMSEMI second source lists 8–20 V VCC (below a 2S pack) – not fitted until its datasheet is read (OQ–13).



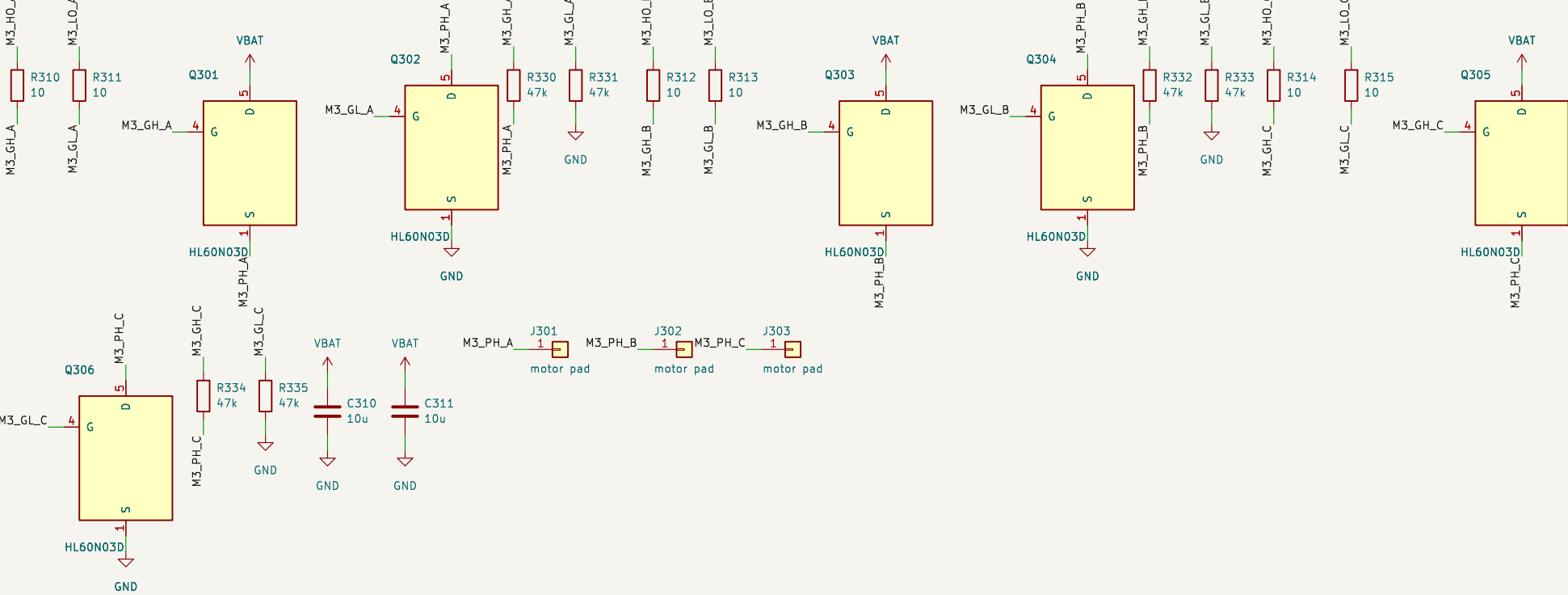
Motor 3 power stage – 6x HL 60N03D (30 V; 4.7 mOhm typ / 6 mOhm max at 10 V gate drive; Qg 34 nC)

Conduction loss at 9.2 A (vendor max current): two FETs conduct, P = 9.2² × 2 × 4.7 mOhm = 0.80 W typ (budgets 4c).

1.0 W at 6 mOhm max. 47 k gate–source resistors hold every FET off while the drivers are unpowered (FD6288

datasheet application circuit R2; D–036 keeps VBAT on the bridge when off). 2x 10 uF 25 V across the bridge;

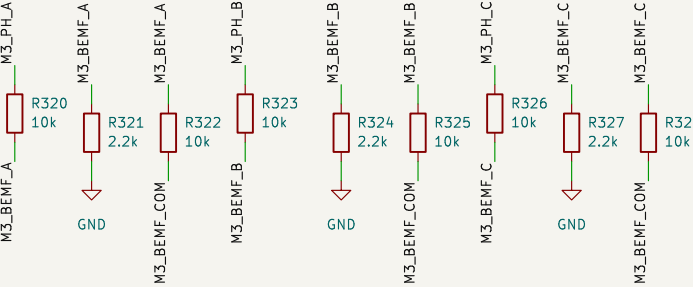
the 100 uF polymer bulk capacitor is on the power sheet.



Motor 3 back–EMF sensing (topology traced from OpenESC–20x20, CERN–OHL–S–2.0; values scaled for 2S)

Each phase: 10 k to the sense node, 2.2 k to GND → 8.8 V × 2.2 / 12.2 = 1.59 V max (OpenESC uses 1 k for 6S).

Each sense node: 10 k to the common (virtual neutral) node on PA1.



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN–OHL–S–2.0)

Sheet: /Motor 3/
File: motor3.kicad_sch

Title: MicroScout 4-in-1 ESC – Motor 3

Size: A3 Date: 2026–10–08

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: G2–draft–1

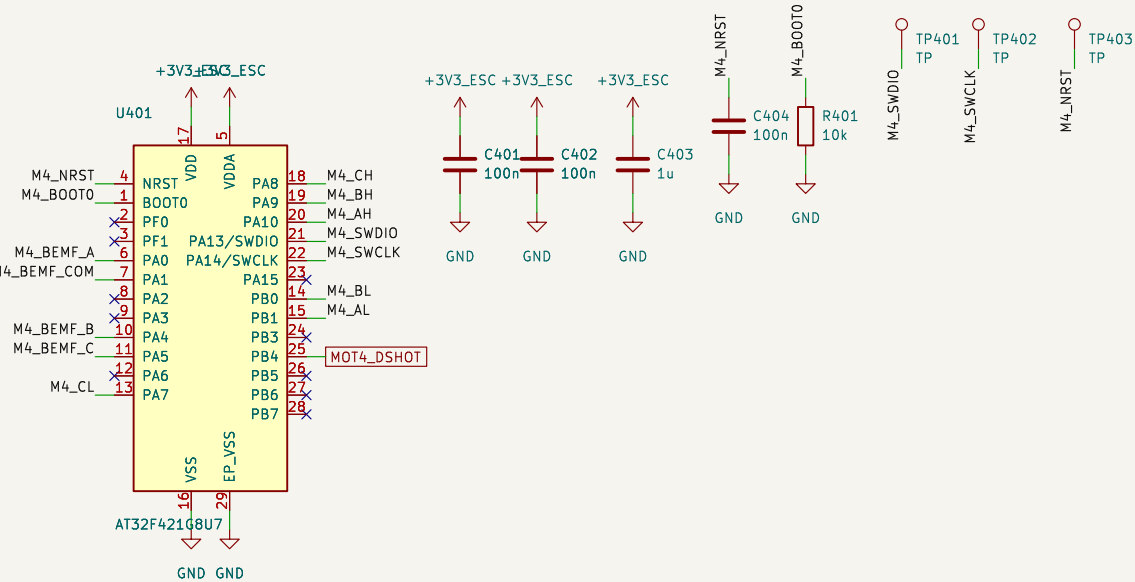
Id: 5/6

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout 4-in-1 ESC – Motor 4

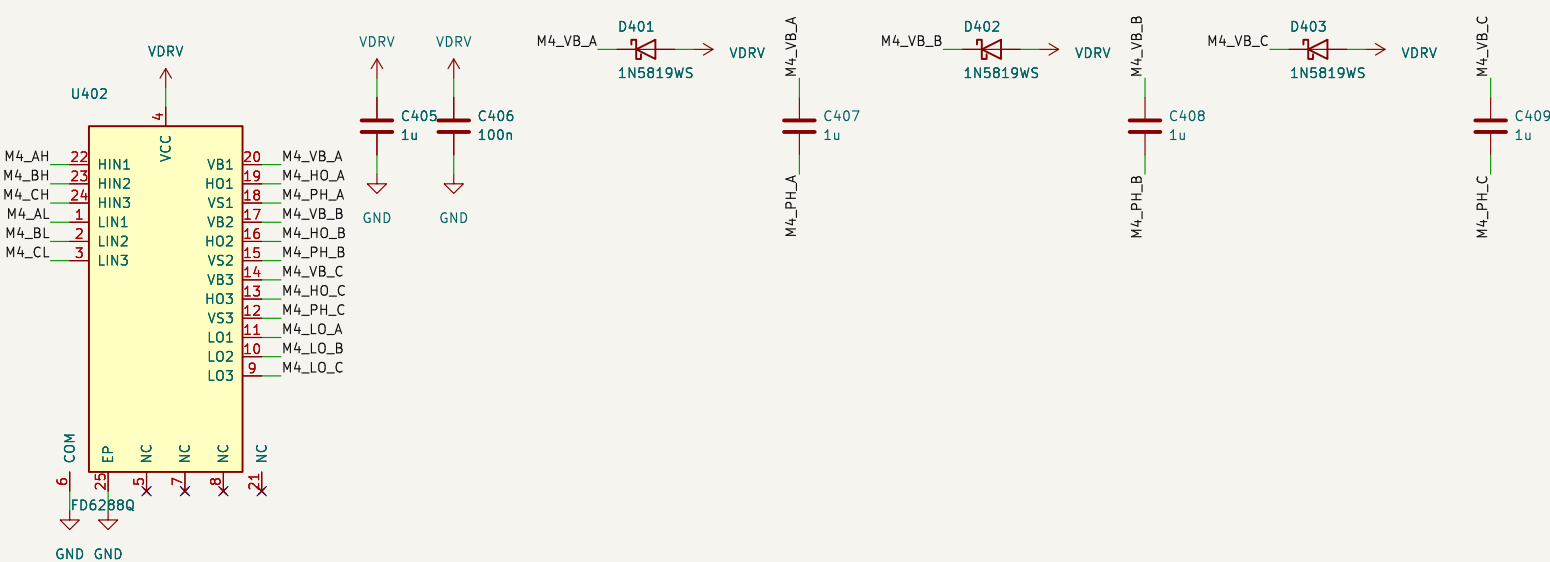
Motor 4 MCU – AT32F421G8U7 with AM32 (pin map: AM32 target OPENESC_20_F421)

PWM: TMR1 CH1–3 + complementary outputs, active high (FD6288 outputs follow inputs; no USE_INVERTED flags).
BEMF comparator inputs PA0/PA4/PA5, neutral PA1. DShot in on PB4 (TMR3 CH1). SWD pads for the first
bootloader flash; later updates over the signal wire. VDD/VDDA decoupling 100 nF each + 1 uF (datasheet Fig. 8;
pin assignment of the values UNCONFIRMED). BOOT0 10 k to GND; NRST 100 nF (value UNCONFIRMED).



Motor 4 gate driver – FD6288Q (Fortior datasheet V1.6 section 1.5)

VCC = VDRV, 9.6 V from the FC boost (D–054); FD6288 VCC 5–20 V, UVLO 4.6 V typ / 5.0 V max rising.
Bootstrap: external Schottky (FD6288 has none) + 1 uF per phase. Cboot >= Qg / dV = 34 nC / 0.1 V = 0.34 uF;
1 uF gives ~34 mV droop per cycle. 10 Ohm gate resistors: agent choice for edge control, tune at G6.
JSMSEMI second source lists 8–20 V VCC (below a 2S pack) – not fitted until its datasheet is read (OQ–13).



Motor 4 power stage – 6x HL 60N03D (30 V; 4.7 mOhm typ / 6 mOhm max at 10 V gate drive; Qg 34 nC)

Conduction loss at 9.2 A (vendor max current): two FETs conduct, P = 9.2² x 2 x 4.7 mOhm = 0.80 W typ (budgets 4c).

1.0 W at 6 mOhm max. 47 k gate–source resistors hold every FET off while the drivers are unpowered (FD6288
datasheet application circuit R2; D–036 keeps VBAT on the bridge when off). 2x 10 uF 25 V across the bridge;
the 100 uF polymer bulk capacitor is on the power sheet.

